

More conditionals

Conditionals with standard booleans:

```
1 should_I_print = True
2 if should_I_print:
3     print("Printing")
```

```
1 if "A".isupper():
2     print("Printing")
```

Conditionals with standard booleans:

```
1 should_l_print = True
2 if should_l_print:
3     print("Printing")
```

Printing

```
1 if "A".isupper():
2     print("Printing")
```

Printing

Conditionals with strings

"" becomes False

other strings become True.

```
1 should_I_print = "Hello"  
2 if should_I_print:  
3     print("Printing")
```

Printing

```
1 should_I_print = ""  
2 if should_I_print:  
3     print("Printing")
```

```
1 if not "":    not false  
2     print("Printing")
```

Printing

Conditionals with numbers

0 and 0.0 become False
other numbers become True.

```
1 should_I_print = 200
2 if should_I_print:
3     print("Printing")
```

Printing

```
1 should_I_print = 0
2 if should_I_print:
3     print("Printing")
```

```
1 if not 0.0:
2     print("Printing")
```

Printing

Conditionals with compound data types

Empty ones, such as [], set(), tuple(), {} become False
other become True.

```
1 should_I_print = {1:10, 2:20, 3:30}
2 if should_I_print:
3     print("Printing")
```

Printing

```
1 should_I_print = []
2 if should_I_print:
3     print("Printing")
```

```
1 if not set():
2     print("Printing")
```

Printing

Conditionals with None

None becomes False
other become True.

```
1 should_I_print = None
2 if should_I_print:
3     print("Printing")
```

```
1 if not None:
2     print("Printing")
```

Printing

Big advantage: Avoids extra checks. (Here if the returned list is empty)

```
1 def get_words_with_e(words):
2     words_with_e = []
3     for word in words:
4         if "e" in word.lower():
5             words_with_e.append(word)
6     return words_with_e
7
8 word_col_1 = ["hej", "tja", "tjena"]
9 word_col_2 = ["a", "b", "c", "d"]
10 for to_test in [word_col_1, word_col_2]:
11     print("\nChecking for e in: " + str(to_test))
12     res = get_words_with_e(to_test)
13     if res: # No need to check if list is empty
14         print("Found the following with 'e': " + str(res))
15     else:
16         print("Found no words containing 'e'")
```

Checking for e in: ['hej', 'tja', 'tjena']

Found the following with 'e': ['hej', 'tjena']

Checking for e in: ['a', 'b', 'c', 'd']

Found no words containing 'e'

This is what it would look like with this check:

```
1 def get_words_with_e(words):
2     words_with_e = []
3     for word in words:
4         if "e" in word.lower():
5             words_with_e.append(word)
6     return words_with_e
7
8 word_col_1 = ["hej", "tja", "tjena"]
9 word_col_2 = ["a", "b", "c", "d"]
10 for to_test in [word_col_1, word_col_2]:
11     print("\nChecking for e in: " + str(to_test))
12     res = get_words_with_e(to_test)
13     if len(res) != 0: # Explicitly check, instead of just write:
14         #if res:
15         print("Found the following with 'e': " + str(res))
16     else:
17         print("Found no words containing 'e'")
```

Checking for e in: ['hej', 'tja', 'tjena']

Found the following with 'e': ['hej', 'tjena']

Checking for e in: ['a', 'b', 'c', 'd']

Found no words containing 'e'

Disadvantage: Opens up for errors if you are not careful

```
1 user_word = input("Type a word: ")
2 last_letter = user_word.lower()[-1]
3
4 if last_letter == "m" or "n":
5     print("Last letter is M or N")
6 else:
7     print("Last letter is NOT M or N")
```

Regardless, what the user writes, this will always result in:

Last letter is M or N

Disadvantage: Opens up for errors if you are not careful

```
1 user_word = input("Type a word: ")
2 last_letter = user_word.lower()[-1]
3
4 if last_letter == "m" or "n":
5     print("Last letter is M or N")
6 else:
7     print("Last letter is NOT M or N")
```

Regardless, what the user writes, this will always result in:

Last letter is M or N

The code should be:

```
1 user_word = input("Type a word: ")
2 last_letter = user_word.lower()[-1]
3
4 if last_letter == "m" or last_letter == "n":
5     print("Last letter IS 'm' or 'n'!")
6 else:
7     print("Last letter is NOT 'm' or 'n'!")
```
